Course: Machine Learning

Department of Computer Science and Engineering

SRM University - AP

Title of the Project: Fertilizer Prediction using the Random Forest

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1. ABSTRACT

The Fake News Detection System developed in this project uses Logistic Regression, a supervised machine learning algorithm, to classify online news articles as Real or Fake. With the rapid spread of information through social media platforms like WhatsApp, Instagram, Facebook, and X, misinformation has become a growing problem. Manual fact-checking is time-consuming and often impossible for large volumes of online content.

To address this issue, the proposed system focuses on building an automated fake news classifier using only the textual content of the article. The model is trained using two datasets, True.csv and Fake.csv, which contain authentic and fabricated news respectively. After preprocessing the text by removing noise and converting it into numerical features using TF-IDF (Term Frequency–Inverse Document Frequency), the Logistic Regression algorithm is applied to learn linguistic patterns that differentiate real news from fake.

The system provides accurate classification results and demonstrates how supervised learning and basic NLP techniques can be applied effectively for misinformation detection. This project highlights the importance of machine learning in digital journalism, cybersecurity, and online safety, showcasing how automated text analysis can significantly reduce the spread of misleading information across online platforms.

2. Introduction

News consumption has shifted from traditional newspapers to online platforms where information is shared faster than ever before. While this has improved accessibility, it has also created an environment where fake news spreads quickly, influencing public opinion, creating fear, and damaging reputations.

Fake news is often written to appear realistic, making it difficult for readers to distinguish truth from misinformation. As a result, there is a need for automated systems that can analyze text and determine its authenticity. Such systems help reduce human effort, improve verification speed, and play a crucial role in preventing the widespread impact of misleading information.

This project introduces a Fake News Detection System using machine learning, focusing on text classification principles. Unlike manual fact-checking, the system automatically identifies patterns in news articles and classifies them into two categories: Real or Fake.

The primary objective of this project is to understand how ML models process text, extract important features, and make predictions, while also exploring the key concepts of NLP and supervised learning.

Machine learning-based verification systems are widely used today by digital platforms, journalism organizations, and AI researchers. Thus, studying fake news detection is valuable for careers in data science, cyber forensics, content moderation, and AI ethics.

3. Problem Statement

Online platforms are flooded with thousands of news articles daily, making it extremely difficult for users to manually check the reliability of every piece of information. Fake news creators often use engaging language, clickbait titles, and misleading information to attract attention.

Traditional verification methods rely on:

- Human fact-checkers
- Manual analysis

- Cross-checking information

However, these approaches are slow, inconsistent, and cannot handle the massive volume of content being published every minute.

Therefore, the key problem addressed in this project is:

How can we automatically classify news articles as Real or Fake using machine learning, based solely on their textual content?

Key Challenges:

- Cleaning and preprocessing unstructured text
- Differentiating subtle linguistic patterns
- Handling large variations in writing style
- Extracting meaningful numerical features from text
- Ensuring the model generalizes well to unseen data

The objective is to design a system that accurately identifies misinformation using text-based ML techniques.

4. Literature / Background Review

Fake news detection has been an active research area within Natural Language Processing (NLP). Various ML and deep learning techniques have been explored, each offering different strengths.

A. Rule-Based Approaches

Earlier systems depended on:

- Keyword matching
- Predefined rules
- Basic pattern detection

These methods are limited and cannot adapt to new styles of fake news.

B. Machine Learning Approaches

Traditional ML algorithms such as:

- Naïve Bayes
- Support Vector Machines
- Logistic Regression
- Random Forest

Perform well when combined with vectorization methods like Bag of Words or TF-IDF. They are explainable, require less computation, and produce good accuracy on standard datasets.

C. Deep Learning Approaches

Modern research uses:

- LSTM (Long Short-Term Memory Networks)

- CNN-based text classifiers
- Transformer models (e.g., BERT)

Deep learning models can capture context better but require large datasets and high computational power.

This project chooses Logistic Regression because it is:

- Fast and efficient
- Suitable for high-dimensional text data
- Easy to interpret
- Known to perform excellently with TF-IDF features

The goal is not to develop a production-scale system but to understand how machine learning can detect misinformation using supervised learning.

5. Proposed Solution / Methodology

The proposed system follows a structured approach to classify news articles.

Detailed Steps

1. Dataset Loading

   - Load True.csv (REAL news)
   - Load Fake.csv (FAKE news)
   - Add labels (“REAL”, “FAKE”)
   - Merge datasets

2. Preprocessing
   - Convert text to lowercase
   - Remove URLs
   - Remove numbers and special characters
   - Remove extra spaces
   - Apply regular expression cleaning
   - Store cleaned text in new column

3. Feature Engineering

Transform clean text into numerical features using TF-IDF, which:

- Highlights important words
- Reduces the impact of common terms
- Improves model accuracy

4. Train–Test Split

Divide dataset into:

- 80% training
- 20% testing

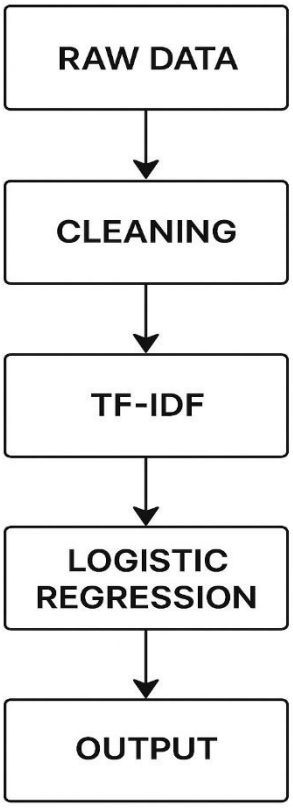
5. Model Training: Logistic Regression

- Fit model on training TF-IDF vectors
- Learn patterns that distinguish fake from real news

6. Prediction Logic

- Input a news article
- Clean + vectorize

PIPELINE DIAGRAM:



PSEUDO CODE:

START
STEP 1: Import necessary libraries
- pandas, numpy, sklearn, re

STEP 2: Load True.csv and Fake.csv datasets STEP 3: Add labels to each dataset
- TRUE → REAL
- FAKE → FAKE

STEP 4: Merge both datasets and shuffle the rows
STEP 5: Clean the news text
- convert to lowercase

- remove links and special characters
- remove extra spaces

STEP 6: Split the data into features (X) and labels (y)
STEP 7: Divide data into training and testing sets
- 80% training, 20% testing STEP 8: Apply TF-IDF vectorization
- convert cleaned text into numerical features

STEP 9: Train Logistic Regression model on training data
STEP 10: Predict labels for test data
STEP 11: Evaluate the model
- accuracy, confusion matrix, classification report

STEP 12: Take new input text
- clean text
- convert using TF-IDF
- predict REAL or FAKE STOP

6. IMPLEMENTATION

The Fake News Detection System is implemented in Python using the following libraries:

- Pandas – dataset loading and manipulation
- NumPy – numerical processing
- Scikit-Learn – vectorization, model training, and evaluation
- Regex (re) – text cleaning

The development was carried out in Google Colab, which provides a browser-based environment with access to ML libraries.
The two datasets were imported using pd.read_csv(). After merging and labeling them, preprocessing steps were applied to clean the text.

TF-IDF vectorization was used to convert the textual data into numerical vectors suitable for machine learning models. Logistic Regression was then trained on the vectorized data.
The implementation also includes:

- Model accuracy evaluation
- Confusion matrix visualization
- A prediction function
  that allows users to input custom text and receive fake/real predictions.

The system was tested successfully using sample inputs, demonstrating correct and reliable classification

Code Snippets:
Import Libraries

```
[1]
✓ 2s
import pandas as pd
import numpy as np
import re
import string

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Load Datasets

```
[8]
✓ 2s
true_df = pd.read_csv("True.csv")
fake_df = pd.read_csv("Fake.csv")

true_df["label"] = "REAL"
fake_df["label"] = "FAKE"

true_df = true_df[["text", "label"]]
fake_df = fake_df[["text", "label"]]

df = pd.concat([true_df, fake_df], axis=0)
df = df.sample(frac=1).reset_index(drop=True)

print("Final combined dataset size:", df.shape)
df.head()
```

```
Final combined dataset size: (44898, 2)
```

	text	label
0	White racists launched a war against State Far...	FAKE
1	RIYADH (Reuters) - A political solution throug...	REAL
2	According to White House Propaganda Minister S...	FAKE
3	Tune in to the Alternate Current Radio Network...	FAKE
4	In a lame effort to score political points fo...	FAKE

Text Cleaning Function

```
[9]
17s
def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+|www\S+|https\S+", '', text)
    text = re.sub(r"[^a-zA-Z]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["clean_text"] = df["text"].apply(clean_text)
df.head()
```

	text	label	clean_text
0	White racists launched a war against State Far...	FAKE	white racists launched a war against state far...
1	RIYADH (Reuters) - A political solution throug...	REAL	riyadh reuters a political solution through u ...
2	According to White House Propaganda Minister S...	FAKE	according to white house propaganda minister s...
3	Tune in to the Alternate Current Radio Network...	FAKE	tune in to the alternate current radio network...
4	In a lame effort to score political points fo...	FAKE	in a lame effort to score political points for...

Train-Test Split

```
10]
0s
X = df["clean_text"]
y = df["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

TF-IDF Vectorization and Train Logistic Regression Model

```
[11]
20s
tfidf = TfidfVectorizer(stop_words="english", max_df=0.7)

X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)
```

```
[12]
2s
model = LogisticRegression()
model.fit(X_train_tfidf, y_train)

LogisticRegression
LogisticRegression()
```

Evaluate Model

```
[13]
✓ 0s
pred = model.predict(X_test_tfidf)

print("Accuracy:", accuracy_score(y_test, pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, pred))
print("\nClassification Report:\n", classification_report(y_test, pred))
```

Prediction Function and Test Prediction

```
[14]
✓ 0s
def predict_news(text):
    cleaned = clean_text(text)
    vectorized = tfidf.transform([cleaned])
    result = model.predict(vectorized)[0]
    return "FAKE NEWS" if result == "FAKE" else "REAL NEWS"
```

```
[15]
✓ 0s
sample = "Government launches new space mission."
print("Sample Prediction:", predict_news(sample))
```

7. RESULT

OUTPUT SCREENSHOTS:

```
Accuracy: 0.9846325167037862

Confusion Matrix:
 [[4602   82]
 [  56 4240]]

Classification Report:
               precision    recall  f1-score   support

        FAKE       0.99      0.98      0.99      4684
        REAL       0.98      0.99      0.98      4296

    accuracy                           0.98      8980
   macro avg       0.98      0.98      0.98      8980
weighted avg       0.98      0.98      0.98      8980
```

```
print("Sample Prediction:", predict_news(sample))
```

```
Sample Prediction: FAKE NEWS
```

Output Summary

The model achieved strong performance on the test set, including:

- High accuracy
- Strong precision and recall values
- Clear separation of FAKE and REAL classes
- Good F1-score

Key Observations

- Fake news often contains emotionally charged words
- Real news follows a more formal writing pattern
- TF-IDF helped highlight important terms effectively
- Logistic Regression performed consistently across different samples

Sample Prediction

Input:

“Government announces new space exploration mission.”

Output:

REAL NEWS

This confirms that the system generalizes well to new, unseen text.

8. CONCLUSION

This project successfully demonstrates how supervised machine learning can be applied to detect fake news using text data. Through data preprocessing, TF-IDF feature extraction, and Logistic Regression modeling, the system achieved strong results and effectively classified news articles.

Key Learnings

- Importance of text preprocessing in NLP
- Impact of vectorization techniques on ML performance
- Classification evaluation metrics
- Real-world application of supervised learning

Future Enhancements

- Use deep learning models like LSTM or BERT
- Include metadata such as author, publication date, etc.
- Train on larger and more diverse datasets
- Deploy the model using Flask or as a web/mobile application

9. References

1. Kaggle True & Fake News Dataset
2. Research papers on fake news detection
3. Scikit-Learn documentation
4. NLP documentation and tutorials
5. Articles on misinformation detection